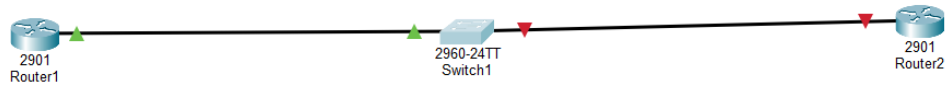


NET512 WEEKLY PRACTICAL

WEEK 5: Cisco Router and Switches Basics



Task 28

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#end
R1#
```

Task 29

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#end
R2#
```

Task 31

```
R1#show ip interface brief


| Interface          | IP-Address | OK? | Method | Status                | Protocol |
|--------------------|------------|-----|--------|-----------------------|----------|
| GigabitEthernet0/0 | 10.10.10.1 | YES | manual | up                    | up       |
| GigabitEthernet0/1 | unassigned | YES | unset  | administratively down | down     |
| Vlan1              | unassigned | YES | unset  | administratively down | down     |


R1#
```

Task 32

```
R2>show ip interface brief


| Interface          | IP-Address | OK? | Method | Status                | Protocol |
|--------------------|------------|-----|--------|-----------------------|----------|
| GigabitEthernet0/0 | 10.10.10.2 | YES | manual | up                    | up       |
| GigabitEthernet0/1 | unassigned | YES | unset  | administratively down | down     |
| Vlan1              | unassigned | YES | unset  | administratively down | down     |


R2>
```

Task 34

```
SW1>ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

SW1>
```

Task 35

```

R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface GigabitEthernet0/0
R1(config-if)#description Link_to_SW1
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

```

Task 36

```

R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID         Local Intrfce   Holdtme    Capability   Platform   Port ID
SW1                Gig 0/0          123        S            2960       Fas 0/1

```

Task 37

```

R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface GigabitEthernet0/0
R1(config-if)#no cdp enable
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

```

Task 38

```

R1#enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no cdp run
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

```

Task 42 and 44

```

SW1#show interface FastEthernet0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 00e0.a3aa.6c02 (bia 00e0.a3aa.6c02)
  Description: Link to R2
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s

```

Task 41

```

SW1#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
FastEthernet0/1    unassigned      YES manual up           up
FastEthernet0/2    unassigned      YES manual administratively down down

```

Task 43

```
SW1#show interface fastethernet0/2
FastEthernet0/2 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 00e0.a3aa.6c02 (bia 00e0.a3aa.6c02)
  Description: Link_to_R2
  BW 100000 Kbit, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 10Mb/s
```

Task 45

```
Keepalive set (10 sec)
Full-duplex, 10Mb/s
```

Task 46

```
R2>show ip interface brief
Interface          IP-Address      OK? Method Status              Protocol
GigabitEthernet0/0 10.10.10.2      YES manual up                  up
GigabitEthernet0/1 unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
R2>
```

Task 47

The symptoms I observed were slow communication, packet loss, collisions, late collisions and interface errors.

Its causing this behaviour because one device is operating in half-duplex while the other is operating in full-duplex.